Introduction

- What is shotcrete?
- providing quality and services to the products

industry since 1979 of shotcrete was

- This innovative technology

introduced to make the work easier and immediate

Shotcrete



Definition

- mortar or high performance concrete conveyed through a hose and pneumatically projected at high velocity onto a backing surface



- An way acceptable of placing cementitious material in a variety of applications.

SHOTCRETE, HIGH PERFORMANCE PRODUCT CONSISTING OF …

Cement + aggregates + water + admixture







+

non-alkaline accelerator

History

- was invented in the early 1900s by American taxidermist Carl Akeley.
- used to fill plaster models of animals.
- In 1911, he was granted a patent.
- Until the 1950's, the wet-mix process was devised, only the dry-mix process was used.



Shotcrete vs. Gunite

- *Shotcrete* is today an all-inclusive term that describes spraying concrete or mortar with either a dry or wet mix process.
- *Gunite* is a trademarked name that is incorrectly used to describe the dry-mix shotcrete process
- Shotcrete emerged as the only acceptable industry term to correctly describe "pneumatically applied concrete".



Reinforcement

- Sprayed concrete is reinforced by conventional steel rods, steel mesh, and/or fibers.
- Fiber reinforcement (steel or synthetic) is also used for stabilization in applications such as slopes or tunneling.



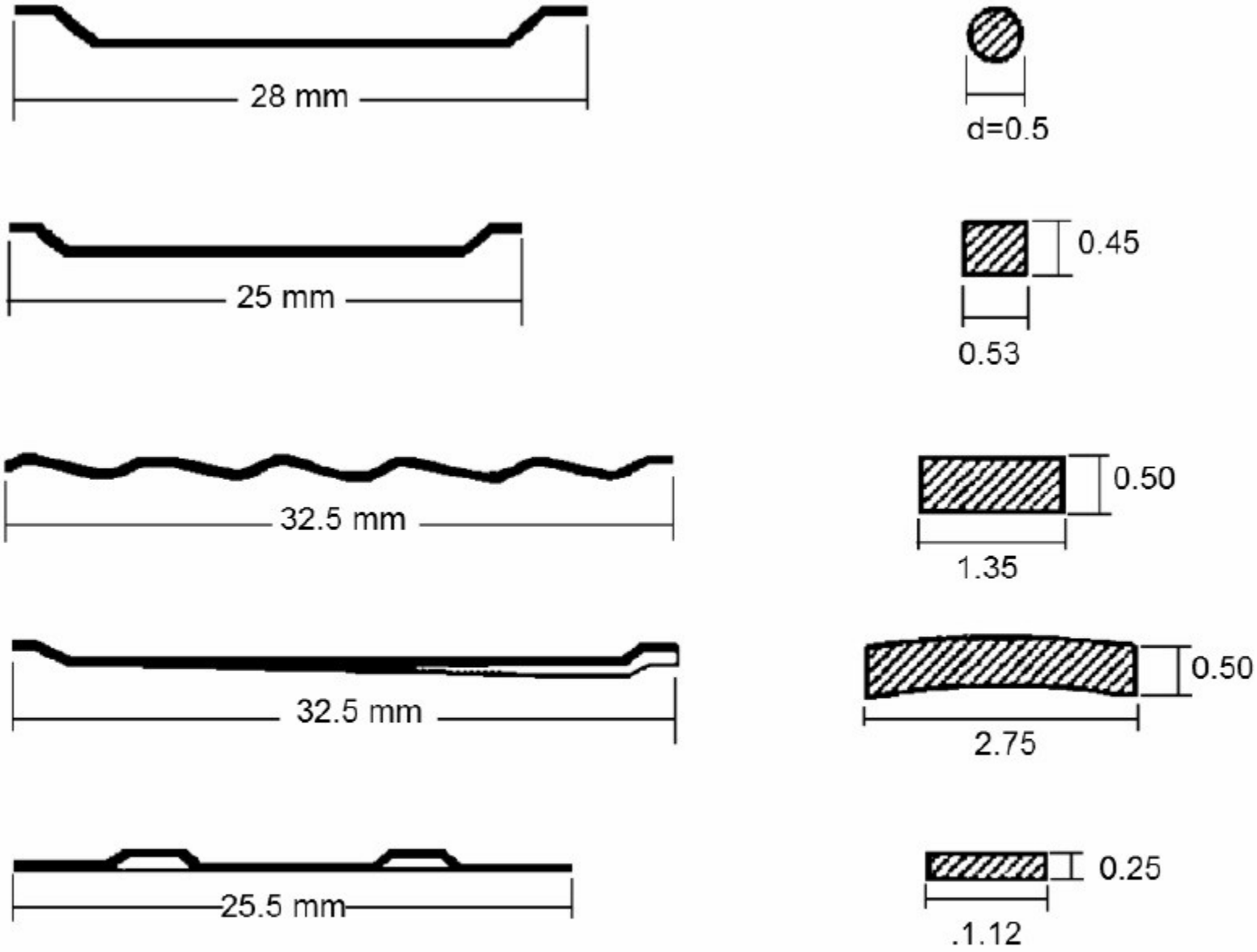


Figure 4. Steel fibre types available on the North American market. After Wood et al (1993). (Note: all dimensions are in mm).

Shotcrete vs. Conventional Concrete

- conventional concrete is first placed and then compacted in the second operation.
- shotcrete undergoes placement and compaction at the same time.
- Shotcrete is more dense, homogeneous, strong, and waterproof .
- It can be impacted onto any type or shape of surface, including vertical or overhead areas

Classification of Shotcrete

1. Dry process
2. Wet process

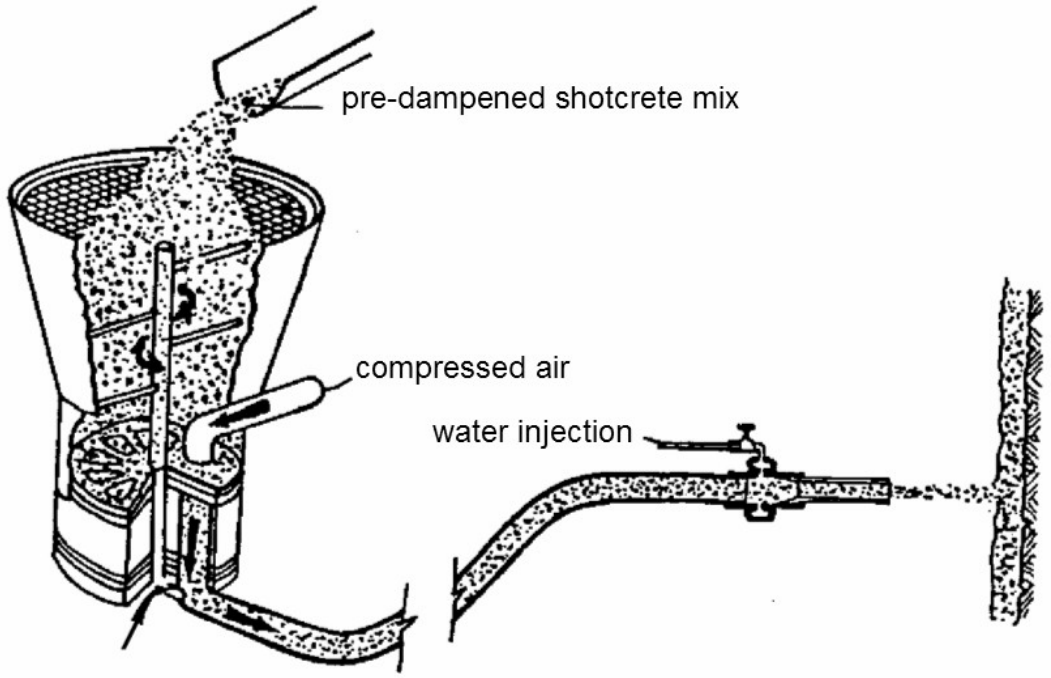
Dry process:

- *Step1:* Pre blended, dry or semi-dampened materials are placed into shotcrete equipment and metered into a hose.
- *Step2:* Compressed air conveys materials at high velocity to the nozzle where the water is added.
- *Step3:* Then the material is consolidated on receiving surface by high impact velocity.



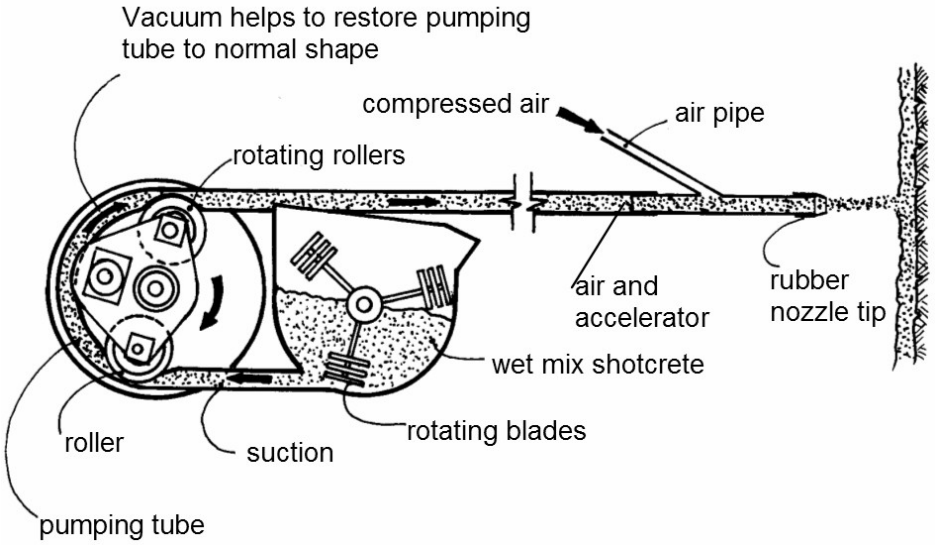
Advantages of Dry process:

- Easy start up, shutdown and clean up.
- Control of materials is on site.
- Nozzle man can be up to 1000ft horizontally or 500ft vertically from the gun.



Wet process:

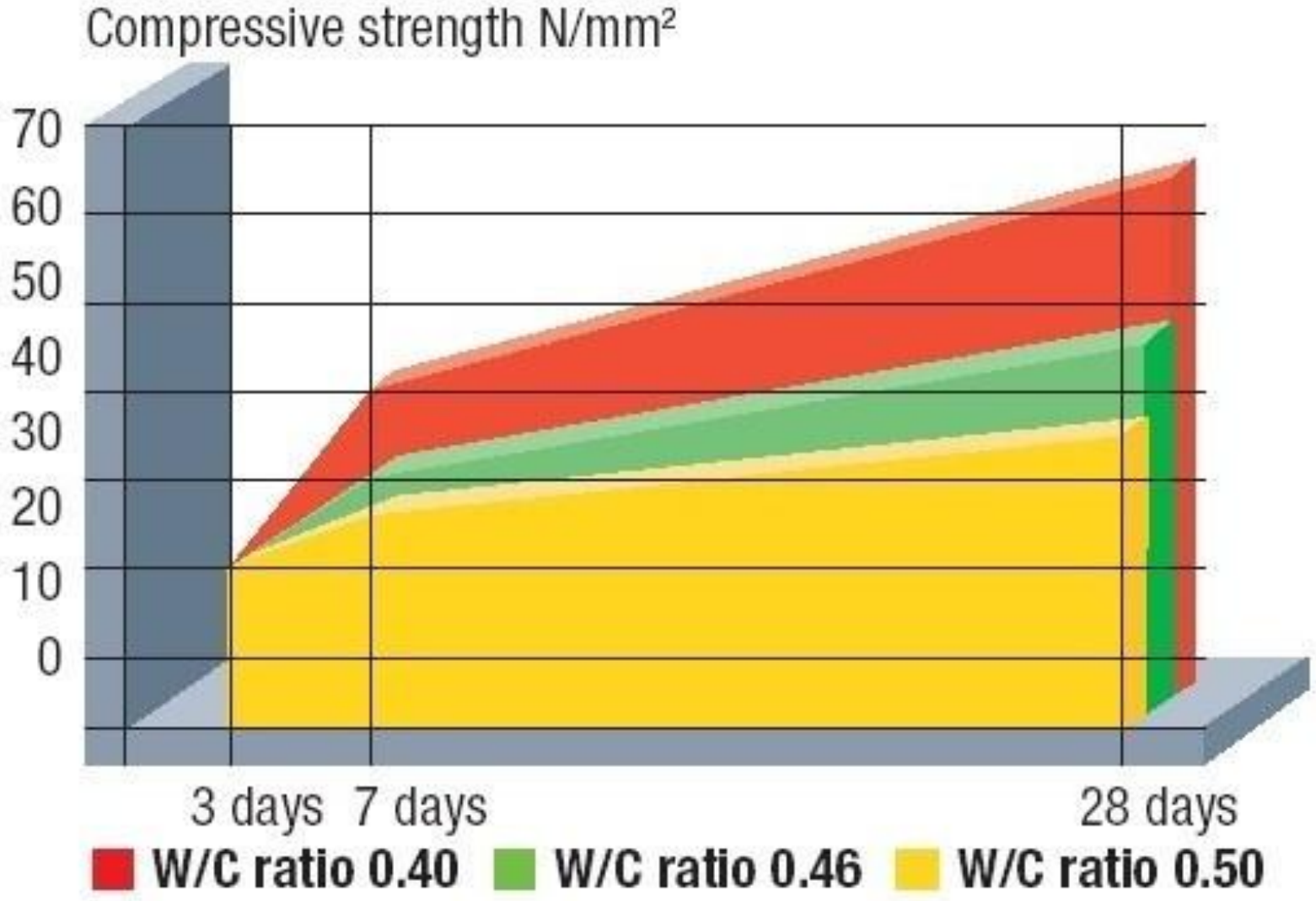
- *Step1:* All ingredients, including water, are thoroughly mixed and introduced into the shotcrete equipment.
- *Step 2:* Wet material is pumped to the nozzle where compressed air is introduced
- *Step 3:* Mostly wet-process shotcreting is done with premixed mortar or small aggregate concrete.



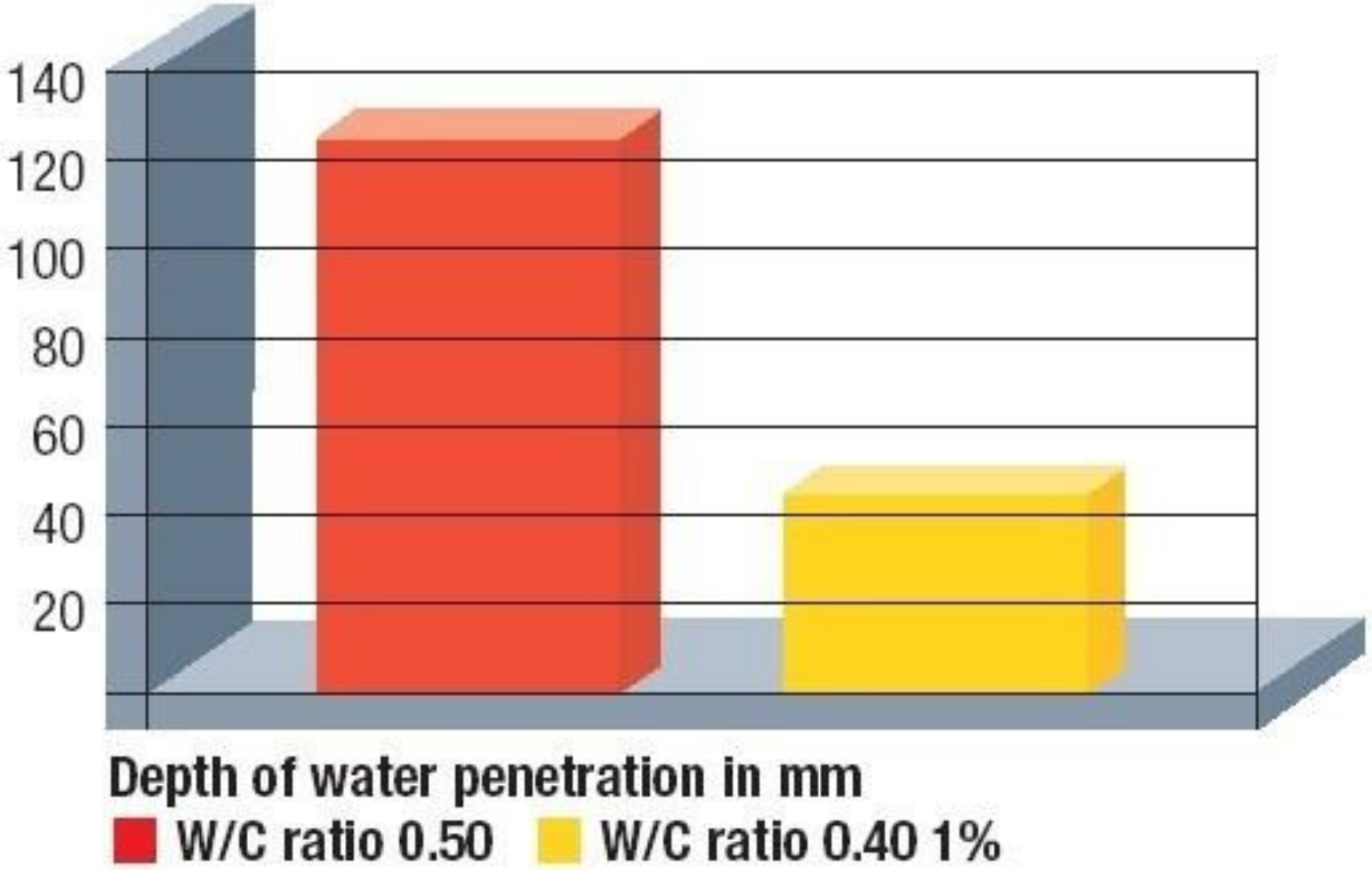
Advantages of Wet process:

- Little or no formwork is required.
- Cost effective method for placing concrete.
- Ideal for irregular surface applications
- Allows for easier material handling in areas with difficult access

Test Results



Effect of W/C ratio on depth of water penetration ❸



Applications of Shotcrete



Rehabilitation of subway tunnels



construction of domed roofs.



Highway culvert repair and arch culvert

CONTENT

- ✓ Definition
- ✓ Introduction
- ✓ Classification
- ✓ Advantages
- ✓ Reinforcement
- ✓ Applications

DEFINITION

“ SHOTCRETE is the generic name for cement, sand and fine aggregate concretes which are applied pneumatically and compacted dynamically under high velocity”.



Shotcrete, high performance product consisting of …

Cement + aggregates + water + admixture







+

non-alkaline accelerator

INTRODUCTION

- Shotcrete technology is providing services to the industry since 1979.
- Later,the development of this technology have been presented by Rose(1985), Frazen (1992) and Morgan(1993).
- This innovative technology of shotcrete was introduced to make the work easier and immediate.
- Shortcreting has proved to be the best method for construction of curved surfaces.
- Domes are now much easier to construct with the advent of with this shotcrete technology.
- Tunnel linings are also becoming easy technology.



Classification of Shotcrete

Classified based on two processes, namely

1. Dry process 2. Wet process

Dry mix shotcrete:

- Step1:Dry shotcrete components, which may be slightly pre dampened to reduce dust, are fed into a hopper with continuous agitation.
- *Step2:* Compressed air is introduced through rotating barrel or feed bowl to convey the materials in a continuous stream through the delivery hose . water is added to the mix at the nozzle.
- *Step3:*Then the material is consolidated on receiving surface by high impact velocity.

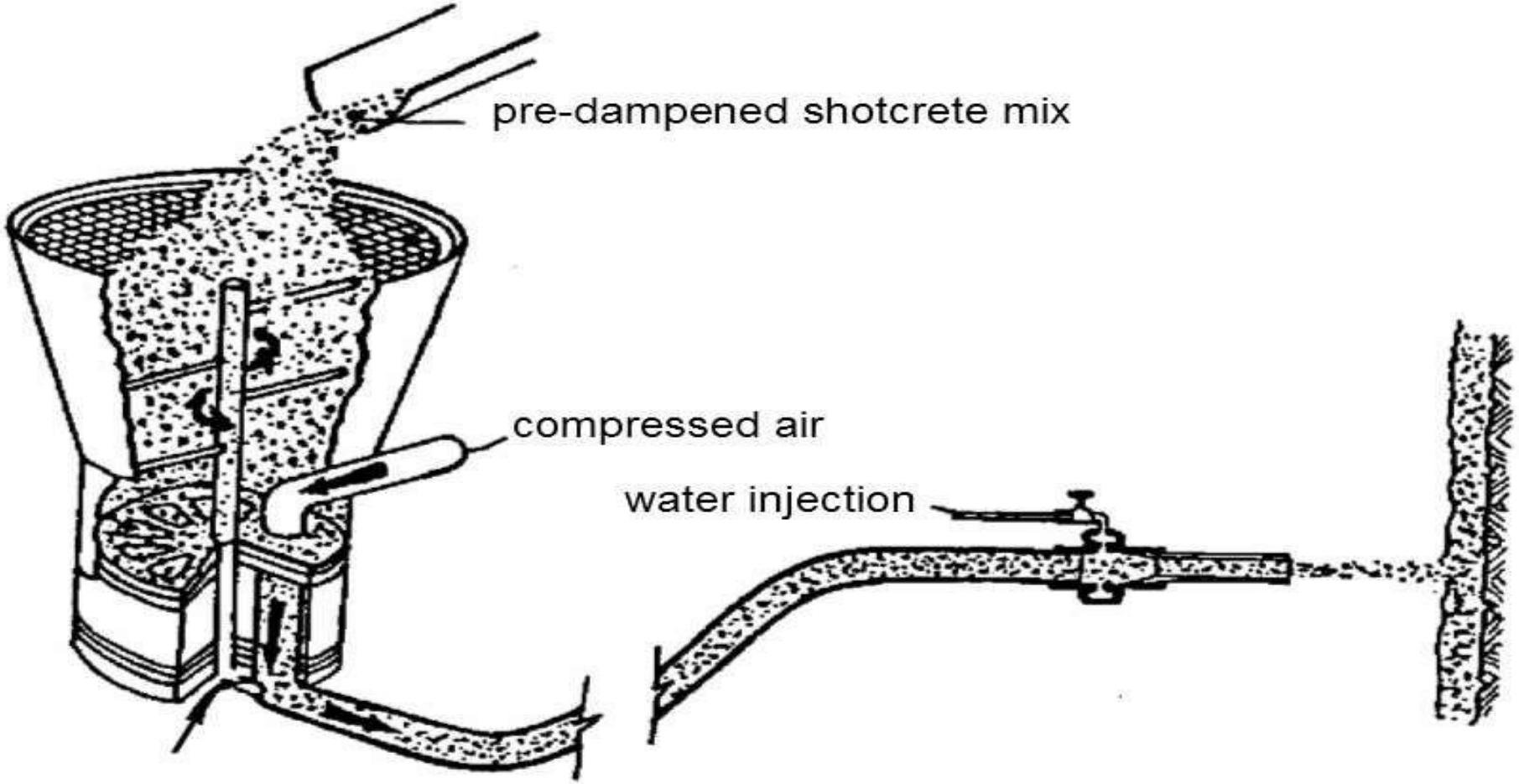


Fig: Simplified sketch of a typical dry mix shotcrete system

Advantages of Dry process:

- Easy start up, shutdown and clean up.
- Control of materials is on site.
- Nozzle man can be up to 1000ft horizontally or 500ft vertically from the gun.
- Widely used in mining



Wet mix shotcrete:

•*Step1:*Shotcrete components and water are mixed(usually in a truck mounted mixer) before delivery into a positive displacement pumping unit.

•*Step 2:* which then delivers the mix hydraulically to the nozzle where air is added to project the material onto the rock surface.

•*Step 3:* Mostly wet-process shotcreting is done with premixed mortar or small aggregate concrete.



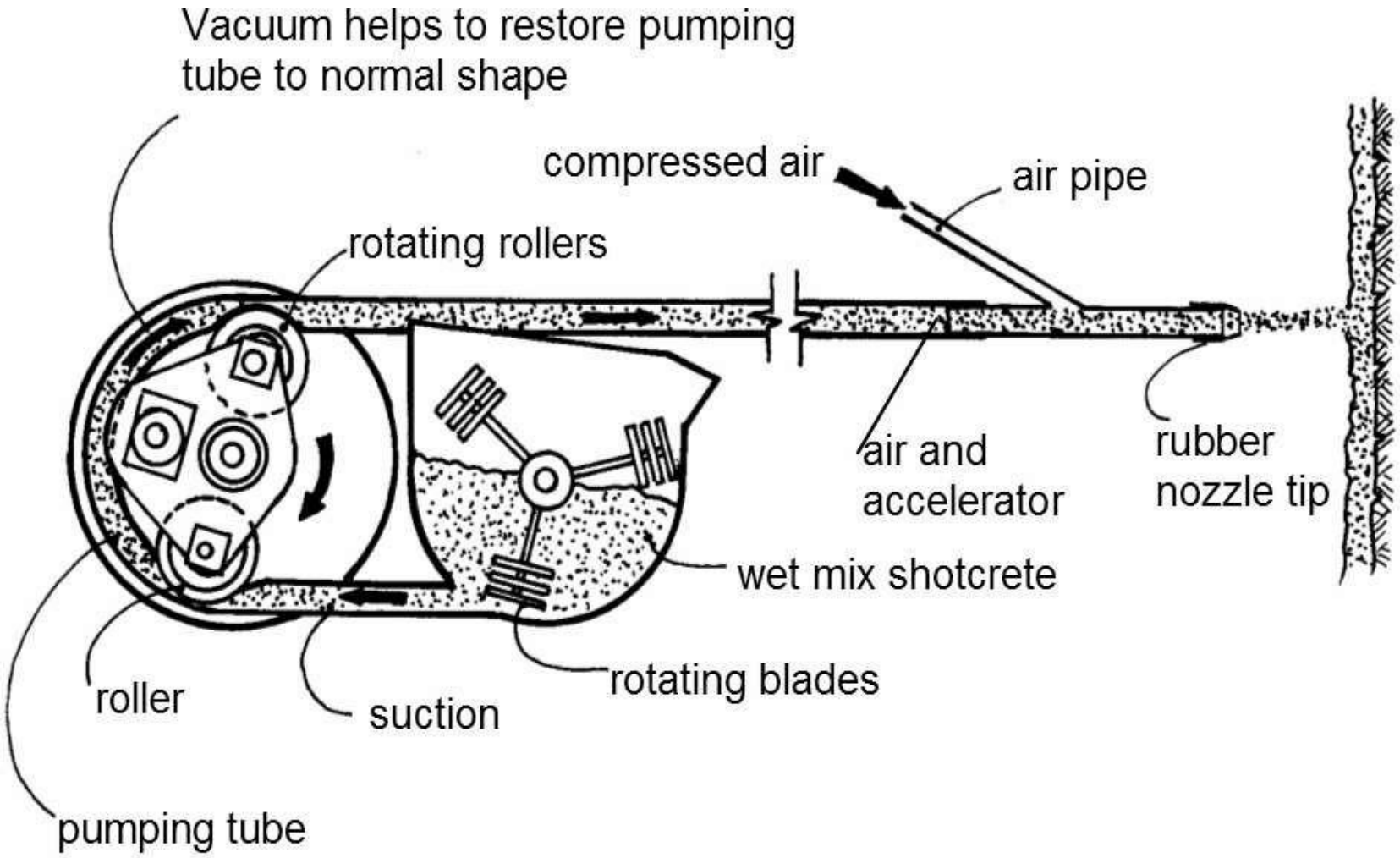


Fig: Typical type of wet mix shotcrete machine

Advantages of Wet process:

- Little or no formwork is required.
- Cost effective method for placing concrete.
- Ideal for irregular surface applications
- Allows for easier material handling in areas with difficult access

Steel fibre reinforced micro silica shotcrete

- Of the many developments in shotcrete technology in recent years, two of the most significant were the introduction of SILICA FUME (used as a cementitious admixture) and STEEL or POLYPROPYLENE FIBRE REINFORCEMENT.
- Silica fume or micro silica is a by-product of the ferro silicon metal industry and is an extremely fine pozzolana.
- Pozzolanas are cementitious materials which react with the calcium hydroxide produced during cement hydration.



Conti...

- Silica fume, added in quantities of 8 to 13% by weight of cement, can allow shotcrete to achieve compressive strengths which are double or triple the value of plain shotcrete mixes.
- The result is an extremely strong, impermeable and durable shotcrete.
- Other benefits include improved flexural strength, improved bond with the rock mass and the ability to place layers of up to 200 mm thick in a single pass because of the shotcrete's 'stickiness'. However, when using wet mix shotcrete, this stickiness decreases the workability of the material and super platicizers are required to restore this workability.

Table.Typical steel fibre reinforced silica fume shotcrete mix designs

COMPONENTS	DRY MIX Kg/m^3	DRY MIX %dry materials	WET MIX Kg/m^3	WET MIX % wet materials
Cement	420	19.0	420	18.1
Silica fume additive	50	2.2	40	1.7
Blended aggregate	1670	75.5	1600	68.9
Steel fibres	60	2.7	60	2.6
Accelerator	13	0.6	13	0.6
Superplasticizer	-	-	6 litres	0.3
Water reducer	-	-	2 litres	0.1
Air entraining admixture	-	-	If required	If required
Water	Controlled at nozzle	Controlled at nozzle	180	7.7

Mesh reinforced shotcrete

- While steel fibre reinforced shotcrete has been widely accepted in both civil and mining engineering, mesh reinforced shotcrete is still widely used and is preferred in some applications.
- In very poor quality, loose rock masses, where adhesion of the shotcrete to the rock surface is poor, the mesh provides a significant amount of reinforcement, even without shotcrete.
- Therefore, when stabilising slopes in very poor quality rock masses or when building bulkheads for underground fill, weldmesh is frequently used to stabilise the surface or to provide reinforcement. In such cases, plain shotcrete is applied later to provide additional support and to protect the mesh against corrosion.

STRENGTH OF SHOTCRETE

Good quality concrete placed by the shotcreting process will substantially exceed 27.6MPa (minimum ACI 318- 08 Building code requirement).

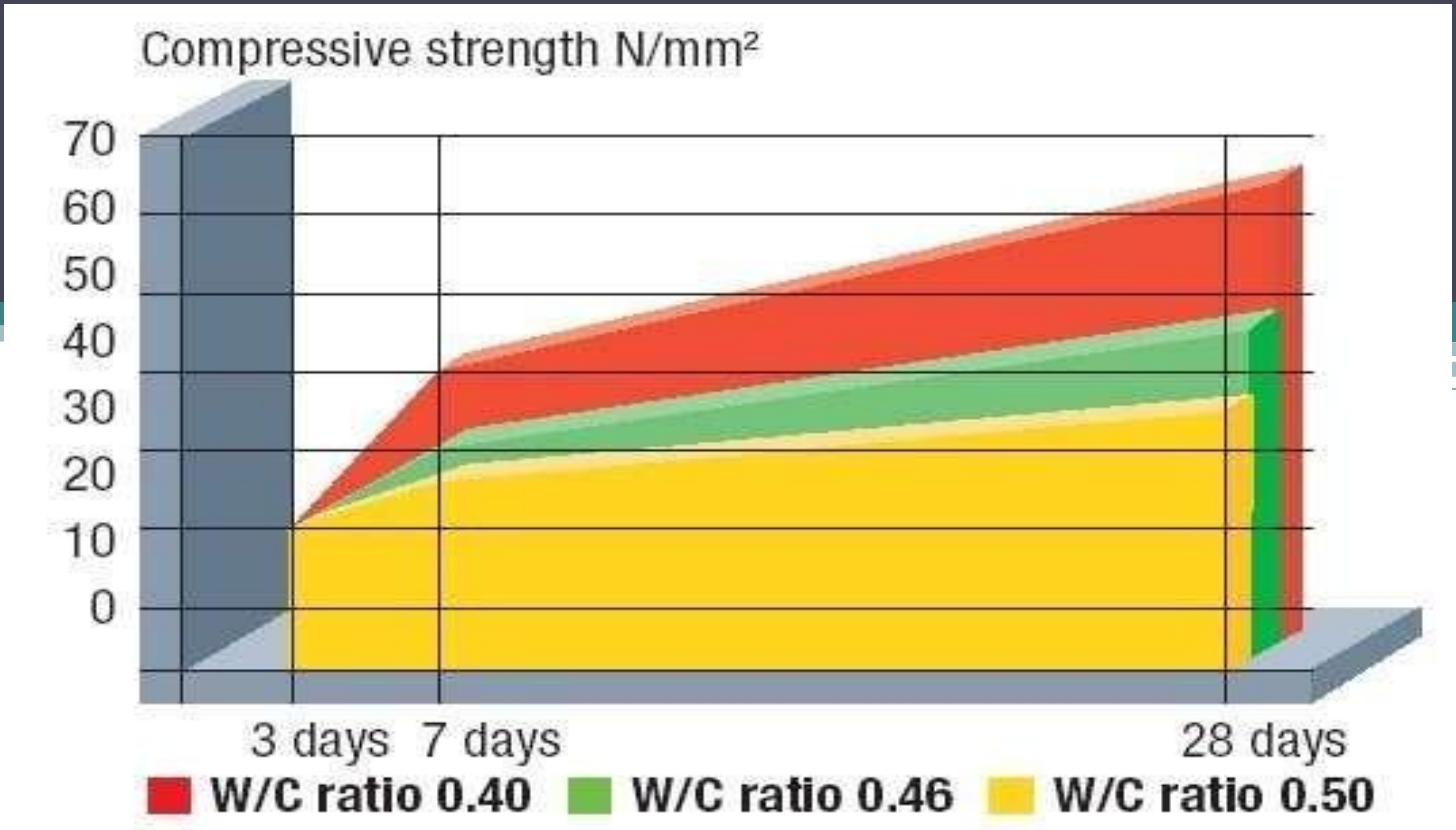


Fig.graph of compressive strength in N/mm^2 v/s time in days

Reinforcement

•Sprayed concrete is reinforced by conventional steel rods, steel mesh or fibers.
•Fiber reinforcement (steel or synthetic) is also used for stabilization in applications such as slopes or tunneling.



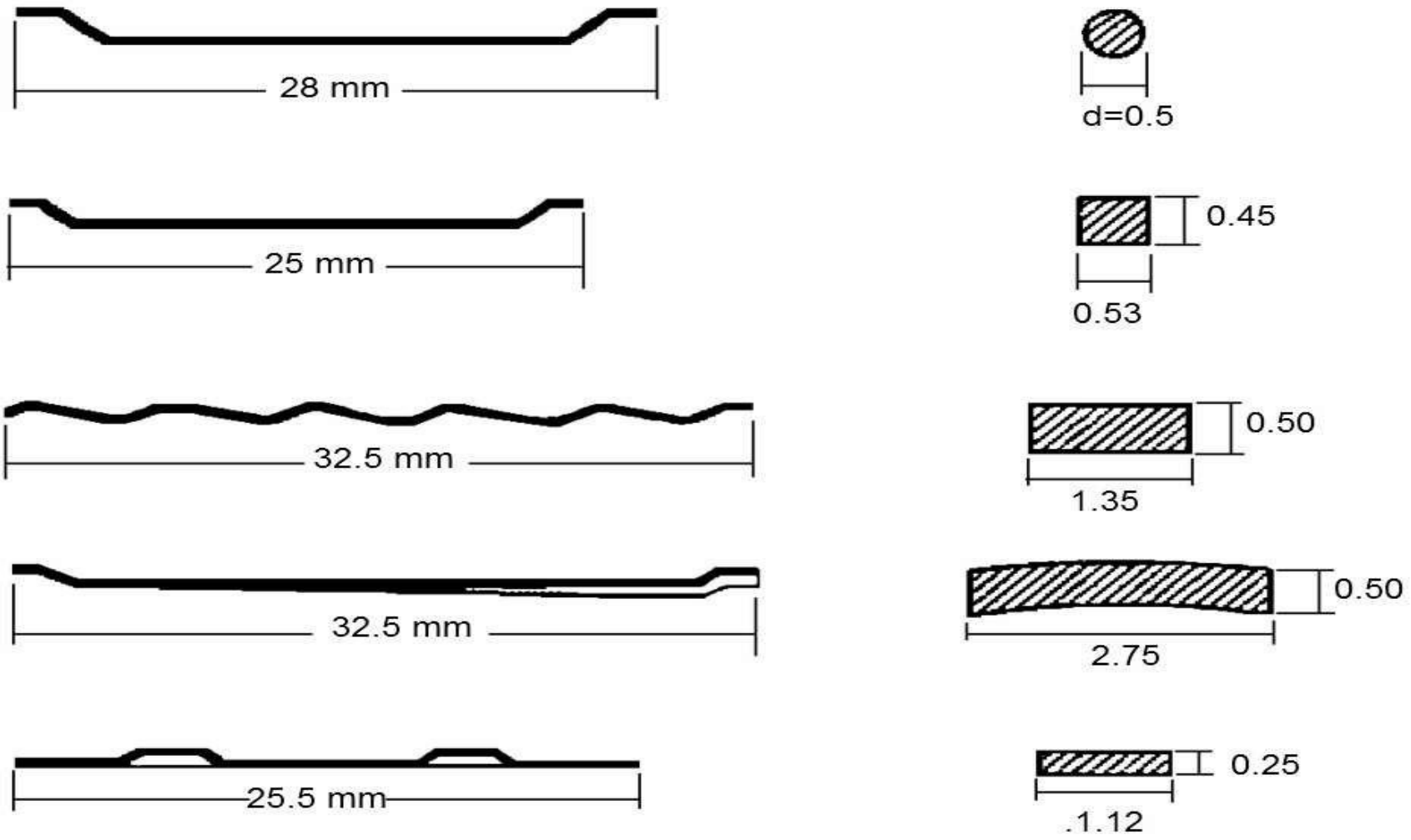


Figure . Steel fibre types available on the North American market. (Note: all dimensions are in mm).

Shotcrete in Building





Figure : A truck mounted shotcrete robot being used in a large civil engineering tunnel. Note that the distance between the nozzle and the rock surface is approximately one metre



Figure : Plastic pipes used to provide drainage for a shotcrete layer applied to a rock mass with water-bearing joints

Shotcrete in Tunneling





Chain link mesh, while very strong and flexible, is not ideal for shotcrete application because it is difficult for the shotcrete to penetrate the mesh.



Welded wire mesh, firmly attached to the rock surface, provides excellent reinforcement for shotcrete

Shotcrete in Retaining wall

